

MALVIKA SAWANT

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EDUCATION

Columbia University

New York, NY

MS in Computer Science, Concentration in Machine Learning, 3.8/4.0

May 2027

Courses: AI, Natural Language Processing, Quantum Computing, Ethical & Responsible AI

Birla Institute of Technology & Sciences, Pilani

Dubai, UAE

B.E. in Computer Science, Minor in Data Science, GPA: 3.7/4.0

Jun 2025

Courses: AI, Machine Learning, Neural Networks, Data Structures & Algos, Operating Systems, Cryptography, Java

Research paper on Neural Networks based Renewable Energy Solutions for Sustainable Homes

Awarded 3rd Place - AI × Cybersecurity CTF, Dubai Police & CTF.ae; won AED 18,000

WORK EXPERIENCE

Columbia University

New York, NY

Research Assistant

Sep 2026 - Present

- Building agentic systems to autonomously compile, deploy, and tune FPGA workloads for Manhattan Reasoning's cloud platform
- Designing authentication and access-control guardrails securing agent access across 10+ shared physical devices

Columbia University

New York, NY

Teaching/Course Assistant - Artificial Intelligence

Sep 2026 - Present

- Instructing and mentoring 100+ students in AI through OH, recitations, grading homework and exams and proctoring assessments

Deriv

Dubai, UAE

Artificial Intelligence Engineer Intern

Jun 2026 - Aug 2026

- Engineered an agentic system autonomously triaged and routed 500+ client escalations/week across Security and Compliance, cutting resolution time by 50%

Deriv

Dubai, UAE

Artificial Intelligence Engineer

Jul 2025 - Jan 2026

- Architected an AI-powered IAM platform leveraging LLMs and agentic workflows, processing 100K+ requests/day with sub-second latency and improving efficiency by 25–40%
- Led development of an AI SOC platform combining ML log filtering and agentic alert analysis; also built fraud-detection systems across 10+ enterprise platforms, reducing false positives by 30%

Deriv

Dubai, UAE

Artificial Intelligence Engineer Intern

Feb 2025 - Jun 2025

- Built an AI tool discovery system using Microsoft AutoGen, enabling company-wide teams to surface and adopt internal AI tools
- Developed anomaly detection models and log-ingestion pipelines integrating Datadog, Okta, and Cloudflare, reducing manual monitoring by 60% and saving ~10 hours/week

Techrobotix FZCO

Dubai, UAE

Digital Solutions Intern

Jun 2023 - Aug 2023

- Deployed a cross-platform campus navigation app using Unity (C#) for Android and iOS, downloaded and utilized by 500+ users

Adiraaj Art of Finance

Mumbai, India

Summer Intern, Finance and Automation

Jun 2022 - Aug 2022

- Automated client alerts using MySQL, maintained databases, and performed financial calculations in Excel

ACADEMIC PROJECTS

- Salary Prediction Web App: Implemented ML model leveraging LinkedIn data; implemented using Python, Pandas, Streamlit
- Attendance Manager: End-to-end hardware-software system using Flutter, MySQL, Arduino
- Data & Systems Projects: JavaFX game development, IPL analytics with Python/MySQL, personal expense tracker

ACTIVITIES

- Professional Assistant Computer Programming, BITS Pilani: Led study groups of 20+ students, assisted with coding labs, grading
- Event Executive, ACM-W: Conducted workshops on Machine Learning, Python programming, and Bootstrap and organized alumni relation events, hosted on-campus quizzes & brainstorming sessions engaging 100+ students per event

SKILLS

AI & ML: Generative AI, ML Security, RAG, Microsoft AutoGen, LangGraph, LangChain, spaCy, Vector Embeddings, Prompt Engineering, Flowise, N8N AI Workflows, scikit learn, Python pandas / NumPy, Langfuse

Design and Development: Unity Technologies, Slack API, Next.js, Node.js, REST APIs, Selenium, Playwright, Firecrawl, Excel

Security Tools: Datadog, OKTA, CVE/Threat Intelligence integrations, Authentication & Authorization, JWT Session Management

Programming Languages: Python, C, Java, C++, C#, JavaScript / TypeScript

Database Management: MySQL, MongoDB, Supabase (PostgreSQL)

Cloud, DevOps & Deployment: Docker, Kubernetes, Google Cloud Platform, AWS, CI/CD best practices, REST APIs

Docker & Kubernetes on O'Reilly, Global Cyber Skills Benchmark CTF 2025: Operation Blackout, by HACKTHEBOX